

Mathematics B

MP–MPI track · Paper no. 3 · Tuesday 14 April 2026

4 hours, no calculator

Worked solutions

Notation follows the paper: $\text{Sp}(M)$ is the real spectrum of M with multiplicities, and $S_f(M) = \frac{1}{n} \sum_{(\lambda, m_\lambda) \in \text{Sp}(M)} m_\lambda f(\lambda)$. As the paper allows, each question may use the results of the previous ones. $\llbracket p, q \rrbracket$ denotes the set of integers between p and q .

Preliminary

Question 1. The sequence satisfies a linear recurrence of order 2 with constant coefficients. Its characteristic equation is $r^2 - \alpha r + 1 = 0$, with discriminant $\Delta = \alpha^2 - 4$. The two roots (in $\mathbb{C}$) have sum α and product 1.

Case $|\alpha| > 2$. There are two distinct real roots $r_\pm = \frac{\alpha \pm \sqrt{\alpha^2 - 4}}{2}$, and $u_n = \lambda r_+^n + \mu r_-^n$. The condition $u_0 = 0$ gives $\mu = -\lambda$, and $u_1 = 1$ gives $\lambda(r_+ - r_-) = 1$ with $r_+ - r_- = \sqrt{\alpha^2 - 4}$. Hence

$$u_n = \frac{1}{\sqrt{\alpha^2 - 4}} \left[\left(\frac{\alpha + \sqrt{\alpha^2 - 4}}{2} \right)^n - \left(\frac{\alpha - \sqrt{\alpha^2 - 4}}{2} \right)^n \right].$$

Case $|\alpha| < 2$. Write $\alpha = 2 \cos \theta$ with $\theta = \arccos(\alpha/2) \in]0, \pi[$. The roots are $e^{\pm i\theta}$ (distinct, complex conjugate), so the real solutions are the sequences $\lambda \cos(n\theta) + \mu \sin(n\theta)$. The condition $u_0 = 0$ gives $\lambda = 0$, and $u_1 = 1$ gives $\mu = 1/\sin \theta$ (note $\sin \theta > 0$):

$$u_n = \frac{\sin(n\theta)}{\sin \theta}, \quad \theta = \arccos \frac{\alpha}{2}.$$

Case $\alpha = 2$. Here 1 is a double root, so $u_n = \lambda + \mu n$, and the initial conditions give $u_n = n$.

Case $\alpha = -2$. Here -1 is a double root, so $u_n = (\lambda + \mu n)(-1)^n$; $u_0 = 0$ gives $\lambda = 0$ and $u_1 = -\mu = 1$ gives $u_n = (-1)^{n+1} n$.

Whenever the two roots $r_+ \neq r_-$ are distinct (real or complex), $u_n = \frac{r_+^n - r_-^n}{r_+ - r_-}$. This is the form used again in question 5c.

Part One

Question 2. The function f is continuous on the segment $[0, 1]$, hence uniformly continuous (Heine's theorem). Let $\omega(h) = \sup\{|f(x) - f(y)| : x, y \in [0, 1], |x - y| \leq h\}$; then $\omega(h) \rightarrow 0$ as $h \rightarrow 0$. Let $R_n = \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right)$. This is a Riemann sum for the regular subdivision of step $1/n$, so $R_n \rightarrow \int_0^1 f(t) dt$.

For $1 \leq k \leq n$,

$$\left| \frac{k}{n+1} - \frac{k}{n} \right| = \frac{k}{n(n+1)} \leq \frac{1}{n+1}, \quad \left| \frac{2k}{2n+1} - \frac{k}{n} \right| = \frac{k}{n(2n+1)} \leq \frac{1}{2n+1}.$$

Hence $|v_n - R_n| \leq \omega\left(\frac{1}{n+1}\right) \rightarrow 0$ and $|w_n - R_n| \leq \omega\left(\frac{1}{2n+1}\right) \rightarrow 0$.

Both sequences converge, and they have the same limit: $\lim v_n = \lim w_n = \int_0^1 f(t) dt$.

Question 3a. *Convergence.* The function f is continuous on $[-2, 2]$, hence bounded by some M . For $x \in [0, 2[$ we have $\sqrt{4-x^2} = \sqrt{2-x}\sqrt{2+x} \geq \sqrt{2}\sqrt{2-x}$, so $\frac{|f(x)|}{\sqrt{4-x^2}} \leq \frac{M}{\sqrt{2}}(2-x)^{-1/2}$. This bound is integrable near 2 (Riemann exponent $\frac{1}{2} < 1$). The same argument works near -2 , so $I(f)$ converges absolutely.

Change of variable. The map $\varphi : \theta \mapsto 2 \cos \theta$ is a C^1 , strictly decreasing bijection from $]0, \pi[$ onto $] -2, 2[$, with $\varphi'(\theta) = -2 \sin \theta$. Moreover $\sqrt{4-4\cos^2\theta} = 2 \sin \theta$ because $\sin \theta > 0$ on $]0, \pi[$. By the change-of-variable theorem for improper integrals (both integrals have the same nature),

$$\int_{-2}^2 \frac{f(x)}{\sqrt{4-x^2}} dx = \int_{\pi}^0 \frac{f(2 \cos \theta)}{2 \sin \theta} (-2 \sin \theta) d\theta = \int_0^{\pi} f(2 \cos \theta) d\theta,$$

so $I(f) = \frac{1}{\pi} \int_0^{\pi} f(2 \cos \theta) d\theta$. The right-hand side is an ordinary integral of a continuous function over a segment.

Question 3b. $I(f_0) = \frac{1}{\pi} \int_0^{\pi} d\theta = 1$, $I(f_1) = \frac{2}{\pi} \int_0^{\pi} \cos \theta d\theta = 0$, $I(f_2) = \frac{1}{\pi} \int_0^{\pi} 4 \cos^2 \theta d\theta = \frac{1}{\pi} \int_0^{\pi} 2(1 + \cos 2\theta) d\theta = 2$.

Question 3c. Expand $(2 \cos \theta)^n = (e^{i\theta} + e^{-i\theta})^n = \sum_{k=0}^n \binom{n}{k} e^{i(2k-n)\theta}$. The function $\theta \mapsto (2 \cos \theta)^n$ is even, so $\int_0^{\pi} = \frac{1}{2} \int_{-\pi}^{\pi}$. For $m \in \mathbb{Z}$ we have $\int_{-\pi}^{\pi} e^{im\theta} d\theta = 2\pi \delta_{m,0}$, so only the term with $2k = n$ survives:

$$\int_0^{\pi} (2 \cos \theta)^n d\theta = \begin{cases} \pi \binom{n}{n/2} & n \text{ even,} \\ 0 & n \text{ odd.} \end{cases}$$

$I(f_n) = \binom{n}{n/2} = \frac{n!}{((n/2)!)^2}$ if n is even, and $I(f_n) = 0$ if n is odd. (Check: this gives 1, 0, 2 for $n = 0, 1, 2$.)

Question 4a. By definition of U_n , $\mathbb{E}\left[f\left(2 \cos \frac{\pi U_n}{n+1}\right)\right] = \frac{1}{n} \sum_{k=1}^n f\left(2 \cos \frac{k\pi}{n+1}\right) = \frac{1}{n} \sum_{k=1}^n g\left(\frac{k}{n+1}\right)$, where $g(t) = f(2 \cos \pi t)$ is continuous on $[0, 1]$. By question 2 (the sequence v_n for g), this tends to $\int_0^1 f(2 \cos \pi t) dt = \frac{1}{\pi} \int_0^{\pi} f(2 \cos \theta) d\theta$, which equals $\frac{1}{\pi} \int_{-2}^2 \frac{f(x)}{\sqrt{4-x^2}} dx$ by 3a.

Question 4b. Let $y \in [-2, 2]$ and $a = \frac{1}{\pi} \arccos \frac{y}{2} \in [0, 1]$. The function $\arccos$ is a strictly decreasing bijection from $[-1, 1]$ onto $[0, \pi]$, and $\frac{k\pi}{n+1} \in]0, \pi[$ for $k \in \llbracket 1, n \rrbracket$. Hence

$$2 \cos \frac{k\pi}{n+1} < y \iff \frac{k\pi}{n+1} > \arccos \frac{y}{2} \iff k > (n+1)a.$$

So the probability is N_n/n , where $N_n = \#\{k \in \llbracket 1, n \rrbracket : k > (n+1)a\}$.

If $(n+1)a \leq n$, then $N_n = n - \lfloor (n+1)a \rfloor$. Otherwise $N_n = 0$ and $n - (n+1)a \in [-1, 0[$. In both cases $|N_n - (n - (n+1)a)| \leq 1$, so $N_n/n \rightarrow 1 - a$. On the other hand, $x \mapsto -\arccos \frac{x}{2}$ is a primitive of $\frac{1}{\sqrt{4-x^2}}$ on $] -2, 2[$, so

$$\frac{1}{\pi} \int_{-2}^y \frac{dx}{\sqrt{4-x^2}} = \frac{1}{\pi} \left(\arccos(-1) - \arccos \frac{y}{2} \right) = 1 - a,$$

which proves the claim.

The same result follows from 4a by squeezing $\mathbf{1}_{]-\infty, y[}$ between continuous functions. The direct count above is shorter.

Part Two

Question 5a. $\chi_2 = \det \begin{pmatrix} X & -1 \\ -1 & X \end{pmatrix} = X^2 - 1$, so $\text{Sp}(T_2) = \{(-1, 1), (1, 1)\}$.

Expanding along the first row, $\chi_3 = X(X^2 - 1) + 1 \cdot \det \begin{pmatrix} -1 & -1 \\ 0 & X \end{pmatrix} = X^3 - X - X = X(X^2 - 2)$, so $\text{Sp}(T_3) = \{(-\sqrt{2}, 1), (0, 1), (\sqrt{2}, 1)\}$.

Question 5b. Expand $\det(XI_n - T_n)$ along its last column. It has two nonzero entries: X in position (n, n) and -1 in position $(n-1, n)$.

- The cofactor of (n, n) is χ_{n-1} .
- Deleting row $n-1$ and column n leaves the block matrix $\begin{pmatrix} XI_{n-2} - T_{n-2} & * \\ 0 & -1 \end{pmatrix}$, whose determinant is $-\chi_{n-2}$. The cofactor sign is $(-1)^{(n-1)+n} = -1$.

Therefore $\chi_n = X\chi_{n-1} + (-1)(-1)(-\chi_{n-2})$, that is,

$$\chi_n = X\chi_{n-1} - \chi_{n-2} \quad \text{for } n \geq 4.$$

With the conventions $\chi_0 = 1$ and $\chi_1 = X$, this relation also holds for $n = 2$ and $n = 3$ (by 5a).

Question 5c. Let $\alpha \in \mathbb{C}$ with $|\alpha| < 2$. Then $\alpha^2 \neq 4$; let δ be a square root of $4 - \alpha^2$, so $\delta \neq 0$. The formula does not depend on which root is chosen: replacing δ by $-\delta$ changes the sign of both the bracket and the prefactor.

Set $r_{\pm} = \frac{\alpha \pm i\delta}{2}$. Then $r_+ + r_- = \alpha$, $r_+ r_- = \frac{\alpha^2 + \delta^2}{4} = 1$ and $r_+ - r_- = i\delta \neq 0$. So $r_{\pm}$ are the two distinct roots of $r^2 = \alpha r - 1$. Define $s_n = \frac{r_+^{n+1} - r_-^{n+1}}{r_+ - r_-}$ (the right-hand side of the formula).

Multiplying $r_{\pm}^2 = \alpha r_{\pm} - 1$ by $r_{\pm}^{n-1}$ gives $s_n = \alpha s_{n-1} - s_{n-2}$. The initial values are

$$s_0 = 1 = \chi_0(\alpha), \quad s_1 = r_+ + r_- = \alpha = \chi_1(\alpha).$$

By 5b, $(\chi_n(\alpha))_n$ satisfies the same recurrence, so $s_n = \chi_n(\alpha)$ for all n by (double) induction:

$$\chi_n(\alpha) = \frac{1}{i\sqrt{4-\alpha^2}} \left[\left(\frac{\alpha + i\sqrt{4-\alpha^2}}{2} \right)^{n+1} - \left(\frac{\alpha - i\sqrt{4-\alpha^2}}{2} \right)^{n+1} \right].$$

(For real α , $\chi_n(\alpha) = u_{n+1}$ with u from question 1.)

Question 5d. By the binomial formula, $(\alpha + i\delta)^{n+1} - (\alpha - i\delta)^{n+1} = \sum_k \binom{n+1}{k} \alpha^{n+1-k} (i\delta)^k (1 - (-1)^k)$. Only odd $k = 2j + 1$ remain, and $(i\delta)^{2j+1} = (-\delta^2)^j (i\delta) = (\alpha^2 - 4)^j i\delta$. Dividing by $2^{n+1}i\delta$ gives

$$\chi_n(\alpha) = \frac{1}{2^n} \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n+1}{2j+1} \alpha^{n-2j} (\alpha^2 - 4)^j.$$

The two sides are polynomials in α that agree on the infinite set $\{|\alpha| < 2\}$, so they are equal as polynomials. Expanding $(X^2 - 4)^j = \sum_{m=0}^j \binom{j}{m} (-4)^m X^{2j-2m}$ and collecting powers gives

$$\chi_n = \sum_{m=0}^{\lfloor n/2 \rfloor} c_{n,m} X^{n-2m}, \quad c_{n,m} = (-1)^m 2^{2m-n} \sum_{j=m}^{\lfloor n/2 \rfloor} \binom{n+1}{2j+1} \binom{j}{m},$$

and all the coefficients of X^{n-2m-1} are zero.

These sums simplify to $c_{n,m} = (-1)^m \binom{n-m}{m}$, which is $\chi_n(X) = U_n(X/2)$ (Chebyshev polynomial of the second kind). Proof: the polynomials $\sum_m (-1)^m \binom{n-m}{m} X^{n-2m}$ equal $X^2 - 1$ and $X^3 - 2X$ for $n = 2, 3$. They also satisfy the recurrence of 5b, by Pascal's rule $\binom{n-1-m}{m} + \binom{n-1-m}{m-1} = \binom{n-m}{m}$.

Question 6. For $\theta \in]0, \pi[$, set $\alpha = 2 \cos \theta \in]-2, 2[$ and take $\delta = 2 \sin \theta$ in 5c, so that $r_{\pm} = e^{\pm i\theta}$. Then

$$\chi_n(2 \cos \theta) = \frac{e^{i(n+1)\theta} - e^{-i(n+1)\theta}}{2i \sin \theta} = \frac{\sin((n+1)\theta)}{\sin \theta}.$$

For $\theta_k = \frac{k\pi}{n+1}$ with $k \in \llbracket 1, n \rrbracket$, we have $\theta_k \in]0, \pi[$ and $\sin((n+1)\theta_k) = 0$, so each $2 \cos \theta_k$ is a root of χ_n . These n values are distinct, because $\cos$ is injective on $]0, \pi[$. Since χ_n is monic of degree n ,

$\chi_n = \prod_{k=1}^n \left(X - 2 \cos \frac{k\pi}{n+1} \right)$: the eigenvalues of T_n are the $2 \cos \frac{k\pi}{n+1}$, $k = 1, \dots, n$, all simple.

Question 7. By question 6, $S_f(T_n) = \frac{1}{n} \sum_{k=1}^n f\left(2 \cos \frac{k\pi}{n+1}\right) = \mathbb{E}\left[f\left(2 \cos \frac{\pi U_n}{n+1}\right)\right]$. By 4a, this tends to $\frac{1}{\pi} \int_{-2}^2 \frac{f(x)}{\sqrt{4-x^2}} dx$.

Question 8a. Since $T_n(a, b, c) = aI_n + T_n(0, b, c)$, $\chi_{T_n(a,b,c)}(X) = \det((X-a)I_n - T_n(0, b, c)) = \chi_{T_n(0,b,c)}(X-a)$. So λ is a root of multiplicity m of the first polynomial if and only if $\lambda - a$ is a root of multiplicity m of the second:

$$\text{Sp}(T_n(a, b, c)) = \{(a + \mu, m) : (\mu, m) \in \text{Sp}(T_n(0, b, c))\}.$$

Question 8b. Let $D_n = \det(XI_n - T_n(0, b, c))$ and expand along the last column as in 5b. The entries $-b$ in position $(n-1, n)$ and $-c$ in position $(n, n-1)$ now appear, and one gets

$$D_n = XD_{n-1} - bc D_{n-2} \quad (n \geq 3), \quad D_1 = X, \quad D_2 = X^2 - bc.$$

The sequence (D_n) therefore depends on (b, c) only through the product bc . The matrix $T_n(0, bc, 1)$ has the same product $bc \cdot 1$, hence the same characteristic polynomial, and so the same spectrum. By 8a,

$$\text{Sp}(T_n(a, b, c)) = \{(a + \mu, m) : (\mu, m) \in \text{Sp}(T_n(0, bc, 1))\}.$$

When $c \neq 0$ this is also a similarity: with $D = \text{diag}(1, c^{-1}, \dots, c^{-(n-1)})$, we get $D T_n(0, b, c) D^{-1} = T_n(0, bc, 1)$.

Question 8c. Assume $bc > 0$ and put $s = \sqrt{bc} > 0$. Then $s \cdot s = bc$, so by 8b the matrices $T_n(0, b, c)$ and $T_n(0, s, s) = s T_n$ have the same characteristic polynomial. By question 6, the eigenvalues of $s T_n$ are $2s \cos \frac{k\pi}{n+1}$. By 8a:

$$\text{The eigenvalues of } T_n(a, b, c) \text{ are } \lambda_k = a + 2\sqrt{bc} \cos \frac{k\pi}{n+1}, \quad k = 1, \dots, n.$$

These are n distinct real numbers, so they are all the complex eigenvalues of $T_n(a, b, c)$, and each is simple.

Question 9a. By 8c, $S_f(T_n(a, b, c)) = \frac{1}{n} \sum_{k=1}^n f\left(a + \sqrt{bc} \cdot 2 \cos \frac{k\pi}{n+1}\right) = S_g(T_n)$, where $g(x) = f(a + \sqrt{bc} x)$ is continuous on $[-2, 2]$. Question 7 gives $\lim_{n \rightarrow \infty} S_f(T_n(a, b, c)) = \frac{1}{\pi} \int_{-2}^2 \frac{f(a + \sqrt{bc} x)}{\sqrt{4-x^2}} dx$.

Question 9b. Let $t = \frac{y-a}{2\sqrt{bc}}$. Then $q_n(y) = \#\{k \in \llbracket 1, n \rrbracket : \cos \frac{k\pi}{n+1} \leq t\}$.

- If $t \geq 1$, i.e. $y \geq a + 2\sqrt{bc}$: every eigenvalue is $\leq y$, so $q_n(y) = n$.
- If $t \leq -1$, i.e. $y \leq a - 2\sqrt{bc}$: $\cos \frac{k\pi}{n+1} > -1 \geq t$, so $q_n(y) = 0$ for every n .

- If $-1 < t < 1$, let $\varphi = \arccos t \in]0, \pi[$. Then $\cos \frac{k\pi}{n+1} \leq t \iff k \geq u := \frac{(n+1)\varphi}{\pi}$. Counting integers as in 4b gives $|q_n(y) - (n - u)| \leq 1$, so $q_n(y)/n \rightarrow 1 - \varphi/\pi > 0$.

For $a - 2\sqrt{bc} < y < a + 2\sqrt{bc}$: $q_n(y) \underset{n \rightarrow \infty}{\sim} \frac{n}{\pi} \arccos\left(\frac{a-y}{2\sqrt{bc}}\right) = \frac{n}{\pi} \int_{-2}^{(y-a)/\sqrt{bc}} \frac{dx}{\sqrt{4-x^2}}$. For $y \geq a + 2\sqrt{bc}$, $q_n(y) = n$; for $y \leq a - 2\sqrt{bc}$, $q_n(y) = 0$.

This agrees with 9a: the fraction of eigenvalues below y follows the arcsine law.

Part Three — Weierstrass's theorem

Recall $Q_n = (1 - X^n)^{2^n}$ and $P_n = Q_n\left(\frac{1-X}{2}\right)$. For $x \in [-1, 1]$ we have $\frac{1-x}{2} \in [0, 1]$, so $0 \leq P_n(x) \leq 1$. This bound is used in question 12.

Question 10a. Let $0 \leq \kappa < \frac{1}{2}$ and $n \geq 1$.

On $[0, \kappa]$. Let $u = x^n \in [0, \kappa^n]$. Bernoulli's inequality $(1 - u)^N \geq 1 - Nu$ (for $u \in [0, 1]$, $N \in \mathbb{N}$) gives

$$1 \geq Q_n(x) = (1 - u)^{2^n} \geq 1 - 2^n \kappa^n = 1 - (2\kappa)^n,$$

so $\sup_{[0, \kappa]} |Q_n - 1| \leq (2\kappa)^n \rightarrow 0$, because $0 \leq 2\kappa < 1$.

On $[1 - \kappa, 1]$. Here $u = x^n \geq (1 - \kappa)^n$. Using $1 - u \leq e^{-u}$,

$$0 \leq Q_n(x) \leq \exp(-2^n u) \leq \exp(-(2(1 - \kappa))^n) \rightarrow 0,$$

because $2(1 - \kappa) > 1$. Both convergences are uniform. (The condition $\kappa < \frac{1}{2}$ is exactly what makes both bounds work.)

Question 10b. Let $0 < \eta \leq 1$ and $\kappa = \frac{1-\eta}{2} \in [0, \frac{1}{2}]$.

- If $x \in [\eta, 1]$, then $\frac{1-x}{2} \in [0, \kappa]$ and $H(x) = 1$. So $|P_n(x) - H(x)| \leq \sup_{[0, \kappa]} |Q_n - 1| \rightarrow 0$.
- If $x \in [-1, -\eta]$, then $\frac{1-x}{2} \in [\frac{1+\eta}{2}, 1] = [1 - \kappa, 1]$ and $H(x) = 0$. So $|P_n(x)| \leq \sup_{[1-\kappa, 1]} Q_n \rightarrow 0$.

Hence $P_n \rightarrow H$ uniformly on $\{x \in [-1, 1] : |x| \geq \eta\}$, which contains $[-1, 1] \setminus [-\eta, \eta]$.

Question 11. The function f is uniformly continuous on $[-1, 1]$, so there is $\delta > 0$ such that $|x - x'| \leq \delta \Rightarrow |f(x) - f(x')| \leq \varepsilon$. Choose $N \geq 1$ with $\frac{2}{N+1} \leq \delta$ and set $c_i = -1 + \frac{2i}{N+1}$ for $0 \leq i \leq N+1$. Then $c_0 = -1$, $c_{N+1} = 1$ and $-1 < c_1 < \dots < c_N < 1$. For $1 \leq i \leq N$, put

$$a_i = f(c_i) - f(c_{i-1}), \quad \text{so that } |a_i| \leq \varepsilon.$$

Let $x \in [-1, 1]$ and let $j \in \llbracket 0, N \rrbracket$ be the largest index with $c_j \leq x$. Then $H(x - c_i) = 1 \iff i \leq j$. Since $f(c_0) = f(-1) = 0$, the sum telescopes:

$$\sum_{i=1}^N a_i H(x - c_i) = \sum_{i=1}^j (f(c_i) - f(c_{i-1})) = f(c_j).$$

As $0 \leq x - c_j \leq c_{j+1} - c_j \leq \delta$, we get $|f(x) - \sum_i a_i H(x - c_i)| = |f(x) - f(c_j)| \leq \varepsilon$.

Question 12. Replacing f by $f - f(-1)$ (and adding the constant back at the end), we may assume $f(-1) = 0$. Apply question 11 with $\varepsilon/3$. This gives $c_1 < \dots < c_N$ in $] -1, 1[$ and $|a_i| \leq \varepsilon/3$ such that $g = \sum_i a_i H(\cdot - c_i)$ satisfies $|f - g| \leq \varepsilon/3$ on $[-1, 1]$.

Domain issue. For $x \in [-1, 1]$, $x - c_i$ ranges over $] -2, 2[$. Outside $[-1, 1]$, P_n does not approximate H ; for instance $P_n(t) \rightarrow +\infty$ for $t < -1$. We therefore rescale. The polynomial $\tilde{P}_n(X) = P_n(X/2)$

satisfies, for $t \in [-2, 2]$: $0 \leq \tilde{P}_n(t) \leq 1$, and $\tilde{P}_n(t) \rightarrow H(t/2) = H(t)$ uniformly on $\{|t| \geq 2\eta\}$ by 10b. We use the polynomial suggested by the hint, with $\tilde{P}_n$ in place of P_n :

$$F_{\varepsilon,n} = \sum_{i=1}^N a_i \tilde{P}_n(X - c_i).$$

Choose $\eta > 0$ such that the intervals $[c_i - 2\eta, c_i + 2\eta]$ are pairwise disjoint. Let $\rho_n = \sup\{|\tilde{P}_n(t) - H(t)| : t \in [-2, 2], |t| \geq 2\eta\}$, so that $\rho_n \rightarrow 0$, and let $A = \sum_i |a_i|$.

Fix $x \in [-1, 1]$. At most one index j satisfies $|x - c_j| < 2\eta$. For that index, bound the term by $|a_j| \cdot 1 \leq \varepsilon/3$, since both $\tilde{P}_n(x - c_j)$ and H lie in $[0, 1]$. For every other index, the term is at most $|a_i| \rho_n$. Hence

$$|F_{\varepsilon,n}(x) - g(x)| \leq \frac{\varepsilon}{3} + A\rho_n,$$

and for n large enough $A\rho_n \leq \varepsilon/3$. Then, on all of $[-1, 1]$, $|f - F_{\varepsilon,n}| \leq |f - g| + |g - F_{\varepsilon,n}| \leq \varepsilon$.

Every continuous function on $[-1, 1]$ is a uniform limit of polynomials (Weierstrass). By an affine change of variable, the same holds on any segment $[\alpha, \beta]$.

The steps of g have height $|a_i| \leq \varepsilon/3$. This is why question 11 requires $a_i \in [-\varepsilon, \varepsilon]$: near each jump, the error of P_n is at most one step.

Part Four — The semicircle law

Question 13a. For every ω , the matrix $X_n(\omega)$ is real symmetric. By the spectral theorem it is orthogonally diagonalisable, $X_n = PDP^\top$, with real eigenvalues $\lambda_1, \dots, \lambda_n$ listed with multiplicity. So its real spectrum accounts for all n eigenvalues. Then $X_n^k = PD^kP^\top$ and $\text{Tr}(X_n^k) = \sum_i \lambda_i^k = \sum_{(\lambda,m) \in \text{Sp}(X_n)} m\lambda^k$, so $S_n(f_k) = \frac{1}{n} \text{Tr}(X_n^k)$, pointwise in ω .

Question 13b. For odd k the integrand is odd, so $\Sigma(f_k) = 0$. For $k = 2p$, substitute $x = 2 \cos \theta$: then $\sqrt{4 - x^2} dx$ becomes $4 \sin^2 \theta d\theta$, and $4 \sin^2 \theta = 4 - (2 \cos \theta)^2$. With the integrals of 3c,

$$\Sigma(f_{2p}) = \frac{1}{2\pi} \int_0^\pi (2 \cos \theta)^{2p} (4 - (2 \cos \theta)^2) d\theta = 2 \binom{2p}{p} - \frac{1}{2} \binom{2p+2}{p+1}.$$

Since $\binom{2p+2}{p+1} = \frac{2(2p+1)}{p+1} \binom{2p}{p}$, this becomes $\binom{2p}{p} (2 - \frac{2p+1}{p+1})$.

$$\Sigma(f_k) = 0 \text{ for odd } k, \text{ and } \Sigma(f_{2p}) = \frac{1}{p+1} \binom{2p}{p} = C_p \text{ (the Catalan numbers } 1, 1, 2, 5, 14, \dots).$$

Question 13c. (H_0) : $\text{Tr}(X_n^0) = n$, so both limits equal $1 = \Sigma(f_0) = \Sigma(f_0)^2$.

(H_1) : $\text{Tr } X_n = n^{-1/2} \sum_i W_{i,i}$, so $\mathbb{E}(\text{Tr } X_n) = 0 = \Sigma(f_1)$. By independence and centring, $\mathbb{E}((\text{Tr } X_n)^2) = \frac{1}{n} \sum_{i,j} \mathbb{E}(W_{i,i} W_{j,j}) = \frac{1}{n} \sum_i \mathbb{E}(W_{i,i}^2) = 1$. Hence $\frac{1}{n^2} \mathbb{E}((\text{Tr } X_n)^2) = \frac{1}{n^2} \rightarrow 0 = \Sigma(f_1)^2$.

(H_2) : $\text{Tr}(X_n^2) = \sum_{i,j} (X_n)_{i,j}^2 = \frac{1}{n} A_n$, where $A_n = \sum_i W_{i,i}^2 + 2 \sum_{i < j} W_{i,j}^2$. Then $\mathbb{E}(A_n) = n + n(n-1) = n^2$, so $\frac{1}{n} \mathbb{E}(\text{Tr } X_n^2) = 1 = \Sigma(f_2)$. The variables $Y_{i,j} = W_{i,j}^2$ ($i \leq j$) are independent, with mean 1 and variance $\sigma^2 = \mathbb{E}(W_{1,1}^4) - 1 < \infty$. So

$$\text{Var}(A_n) = \sigma^2(n + 4 \cdot \frac{n(n-1)}{2}) = \sigma^2(2n^2 - n), \quad \mathbb{E}(A_n^2) = n^4 + \sigma^2(2n^2 - n).$$

Hence $\frac{1}{n^2} \mathbb{E}((\text{Tr } X_n^2)^2) = \frac{\mathbb{E}(A_n^2)}{n^4} \rightarrow 1 = \Sigma(f_2)^2$.

Question 14. For every real x , $0 \leq g_{k,B}(x) \leq \frac{x^{2k}}{B^k}$. Indeed, if $|x| \leq B$ the left side is 0; if $|x| > B$, then $|x|^{2k} = |x|^k |x|^k \geq |x|^k B^k$. The map $h \mapsto S_n(h)$ is linear and nondecreasing (an average with

nonnegative weights), so $0 \leq S_n(g_{k,B}) \leq S_n(f_{2k})/B^k$. Markov's inequality then gives

$$\mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) \leq \frac{\mathbb{E}(S_n(g_{k,B}))}{\varepsilon} \leq \frac{\mathbb{E}(S_n(f_{2k}))}{\varepsilon B^k}.$$

The expectation on the right is finite: $\text{Tr}(X_n^{2k})$ is a polynomial in the $W_{i,j}$, which have finite moments of every order.

Question 15. Let $k' \geq k$. For $|x| > B > 1$ we have $|x|^k \leq |x|^{k'}$, so $g_{k,B} \leq g_{k',B}$. Then, by question 14,

$$\mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) \leq \mathbb{P}(S_n(g_{k',B}) \geq \varepsilon) \leq \frac{\mathbb{E}(S_n(f_{2k'}))}{\varepsilon B^{k'}}.$$

By 13a and $(H_{2k'})$, $\mathbb{E}(S_n(f_{2k'})) = \frac{1}{n} \mathbb{E}(\text{Tr } X_n^{2k'}) \rightarrow \Sigma(f_{2k'}) \leq 4^{k'}$, because $x^{2k'} \leq 4^{k'}$ on $[-2, 2]$ and $\Sigma(f_0) = 1$. Hence, for every $k' \geq k$,

$$\limsup_{n \rightarrow \infty} \mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) \leq \frac{1}{\varepsilon} \left(\frac{4}{B} \right)^{k'}.$$

Letting $k' \rightarrow \infty$ (with $4/B < 1$) gives $\lim_n \mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) = 0$.

Question 16. By the Bienaymé–Chebyshev inequality,

$$\mathbb{P}(|S_n(f_k) - \mathbb{E}S_n(f_k)| \geq \varepsilon) \leq \frac{\text{Var}(S_n(f_k))}{\varepsilon^2},$$

and by (H_k) ,

$$\text{Var}(S_n(f_k)) = \frac{1}{n^2} \mathbb{E}((\text{Tr } X_n^k)^2) - \left(\frac{1}{n} \mathbb{E}(\text{Tr } X_n^k) \right)^2 \longrightarrow \Sigma(f_k)^2 - \Sigma(f_k)^2 = 0.$$

Question 17a. Weierstrass's theorem (Part Three) on $[-B, B]$ gives a polynomial $P = \sum_{k=0}^d p_k X^k$ with $\sup_{[-B, B]} |f - P| \leq \varepsilon/6$. By linearity, 13a and the (H_k) , $\mathbb{E}(S_n(P)) = \sum_k p_k \frac{1}{n} \mathbb{E}(\text{Tr } X_n^k) \rightarrow \sum_k p_k \Sigma(f_k) = \Sigma(P)$. So there is an integer N such that $|\mathbb{E}(S_n(P)) - \Sigma(P)| \leq \varepsilon/6$ for all $n \geq N$.

Since $f = f \mathbf{1}_{|x| \leq B}$, linearity of S_n gives the identity

$$\begin{aligned} S_n(f) - \Sigma(f) &= S_n((f - P)\mathbf{1}_{|x| \leq B}) - S_n(P\mathbf{1}_{|x| > B}) + [S_n(P) - \mathbb{E}S_n(P)] \\ &\quad + [\mathbb{E}S_n(P) - \Sigma(P)] + [\Sigma(P) - \Sigma(f)]. \end{aligned}$$

Three of these terms are small:

- $|S_n((f - P)\mathbf{1}_{|x| \leq B})| \leq \varepsilon/6$, since it is an average of values bounded by $\varepsilon/6$;
- $|\Sigma(P) - \Sigma(f)| \leq \frac{\varepsilon}{6} \Sigma(f_0) = \frac{\varepsilon}{6}$, since $[-2, 2] \subset [-B, B]$;
- $|\mathbb{E}S_n(P) - \Sigma(P)| \leq \varepsilon/6$ for $n \geq N$.

So for $n \geq N$, if $|S_n(f) - \Sigma(f)| \geq \varepsilon$, then $|S_n(P\mathbf{1}_{|x| > B})| + |S_n(P) - \mathbb{E}S_n(P)| \geq \varepsilon/2$, and one of the two terms is at least $\varepsilon/4$. The union bound gives

$$\mathbb{P}(|S_n(f) - \Sigma(f)| \geq \varepsilon) \leq \mathbb{P}(|S_n(P\mathbf{1}_{|x| > B})| \geq \frac{\varepsilon}{4}) + \mathbb{P}(|S_n(P) - \mathbb{E}(S_n(P))| \geq \frac{\varepsilon}{4}).$$

Question 17b. Let $K = \{k \leq d : p_k \neq 0\}$ and $\varepsilon_k = \frac{\varepsilon}{4(d+1)|p_k|}$ for $k \in K$.

- Since $|P(x)|\mathbf{1}_{|x| > B} \leq \sum_k |p_k| g_{k,B}(x)$, we have $|S_n(P\mathbf{1}_{|x| > B})| \leq \sum_k |p_k| S_n(g_{k,B})$. If this sum is at least $\varepsilon/4$, then some $k \in K$ has $S_n(g_{k,B}) \geq \varepsilon_k$. So $\mathbb{P}(|S_n(P\mathbf{1}_{|x| > B})| \geq \varepsilon/4) \leq \sum_{k \in K} \mathbb{P}(S_n(g_{k,B}) \geq \varepsilon_k)$, and each term tends to 0 by question 15.
- Likewise $S_n(P) - \mathbb{E}S_n(P) = \sum_k p_k (S_n(f_k) - \mathbb{E}S_n(f_k))$, so $\mathbb{P}(|S_n(P) - \mathbb{E}S_n(P)| \geq \varepsilon/4) \leq \sum_{k \in K} \mathbb{P}(|S_n(f_k) - \mathbb{E}S_n(f_k)| \geq \varepsilon_k)$, and each term tends to 0 by question 16.

With 17a:

$\lim_{n \rightarrow \infty} \mathbb{P}(|S_n(f) - \Sigma(f)| \geq \varepsilon) = 0$: the empirical spectral distribution of X_n converges in probability to the semicircle law $\frac{1}{2\pi} \sqrt{4 - x^2} dx$ on $[-2, 2]$ (Wigner's theorem).

Compare with Parts One and Two: for the deterministic matrices T_n the limit was the arcsine law $\frac{dx}{\pi \sqrt{4 - x^2}}$. For random symmetric matrices it is the semicircle.

Unofficial worked solutions. The questions are those of the X-ESPCI 2026 Mathematics B paper (MP-MPI track); the original French paper is in this folder.